Data Science Practicum II, Proposal

Prakash Ghatani
Predicting insurance claim costs
Regis University
Regis University, Denver, CO, USA
pghatani@regis.edu

Abstract

This project applies machine learning to predict insurance claim costs using the Allstate Claims Severity dataset from Kaggle. The dataset contains 188,318 insurance claims with categorical and numerical features, and the target variable, *loss*, represents the dollar amount of each claim. The business goal is to help insurance claims managers identify potentially high-cost claims earlier, improve reserve planning, prioritize claim reviews, and allocate resources more effectively.

Exploratory data analysis showed that claim costs are strongly right-skewed. Approximately 83.59% of claims are below $5,000, while only 23 claims exceed $40,000, showing that extremely high-cost claims are rare.

Three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Before tuning, Random Forest achieved the best performance with an RMSE of $1,903.31, compared with $1,945.67 for Gradient Boosting and $2,016.61 for Linear Regression.

After hyperparameter tuning, Gradient Boosting achieved the best overall performance, with an RMSE of $1,884.82, an MAE of $1,218.61, and an R^2 of 0.5646. Based on these results, Gradient Boosting is recommended as a decision-support tool for ranking and prioritizing claims by predicted cost.

Insurance executives should pilot the model by routing the top 5% of model-scored claims to experienced adjusters for early review before reserves are finalized. The 5% threshold should be treated as an initial pilot threshold and validated before full deployment because extreme high-cost claims remain more difficult to predict accurately.

I. Introduction/Background

Insurance companies process many claims every day, and the final cost of each claim can vary widely. Identifying potentially expensive claims early is important because it can help claims managers prioritize reviews, allocate resources more effectively, and improve reserve planning. This project focuses on using machine learning to predict insurance claim costs and provide decision support to insurance claims professionals. The target variable, *loss*, represents the dollar amount of each insurance claim.

The data used in this project comes from the Allstate Claims Severity competition published on Kaggle in 2016. The dataset contains 188,318 claims and 132 columns, including 116 categorical features, 14 numerical features, an ID column, and the target variable *loss*. The data was imported into Python and examined using tools such as pandas and scikit-learn. Initial inspection found no duplicate rows or missing values. Because the target variable is a continuous dollar amount, this problem is appropriate for regression analysis.

Exploratory data analysis showed that claim costs are strongly right-skewed. Most claims are relatively inexpensive, while a small number of claims have extremely high costs. Approximately 83.59% of claims are below $5,000, while only 23 out of 188,318 claims exceed $40,000. Although these high-cost claims are rare, they can create greater financial risk. Identifying potentially expensive claims earlier may allow claims managers to review higher-risk cases sooner and make better-informed decisions when planning reserves.

Three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Before tuning, Random Forest achieved the best performance with an RMSE of $1,903.31, compared with $1,945.67 for Gradient Boosting and $2,016.61 for Linear Regression. After hyperparameter tuning, Gradient Boosting achieved the best overall performance, with an RMSE of $1,884.82, an MAE of

$1,218.61, and an R^2 of 0.5646. These results show that the model can provide useful claim-cost predictions, although extreme high-cost claims remain more difficult to predict accurately.

Because of these limitations, the model should be used to support claims professionals rather than replace their judgment. The dataset contains anonymized features and does not include information such as claim dates, detailed policy information, or external factors, which limits business interpretation and generalizability. Therefore, the recommended approach is to use the tuned Gradient Boosting model to rank claims by predicted cost and initially route the top 5% of scored claims to experienced adjusters for early review before reserves are finalized. The 5% threshold should be treated as an initial pilot threshold and validated before full operational deployment.

‘‘‘latex

II. Problem Statement

Insurance companies need to estimate claim costs accurately to set appropriate reserves and manage financial resources effectively. However, claim costs can vary widely, and a small number of claims can become extremely expensive. In the Allstate Claims Severity dataset, approximately 83.59% of claims are below $5,000, while only 23 out of 188,318 claims exceed $40,000. This demonstrates that very high-cost claims are rare but may create greater financial risk for an insurance company.

The main problem addressed in this project is how machine learning can be used to predict insurance claim costs and identify potentially expensive claims early enough to support claims-management decisions. The target variable, *loss*, represents the dollar amount of each insurance claim. Predicting claim costs can help claims managers prioritize claim reviews, allocate resources more effectively, and support better reserve planning.

Predicting claim costs is challenging because the distribution of *loss* is strongly right-skewed. Most claims have relatively low costs, while a small number have extremely high costs. In addition, the dataset contains anonymized predictors and does not include claim dates, detailed policy information, or external factors. These limitations reduce the ability to interpret individual predictors and make extreme high-cost claims more difficult to predict accurately. Therefore, the model is intended to support claims professionals rather than replace their professional judgment.

To address this problem, three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Their predictive performance was compared using evaluation metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). The best-performing model was then selected to rank claims according to predicted cost.

The final business objective is to use the selected model as a decision-support tool and conduct an initial pilot in which the top 5% of model-scored claims are routed to experienced adjusters for early review before reserves are finalized. The 5% threshold is intended as a starting point and should be validated to determine how effectively it captures high-cost claims before full operational deployment. ‘‘‘

III. Related Work

Machine learning provides several approaches for predicting continuous outcomes such as insurance claim costs. Regression and tree-based ensemble methods are commonly used for this type of prediction because they can model relationships between input features and a continuous target variable. In this project, Linear Regression, Random Forest, and Gradient Boosting were evaluated to determine which approach could best predict the *loss* variable in the Allstate Claims Severity dataset. The dataset, published through Kaggle, contains 188,318 claims with 116 categorical and 14 numerical features used for prediction.

Linear Regression was selected as the baseline model because the target variable is a continuous dollar amount. A baseline model provides a simple reference point for evaluating whether more advanced machine-learning methods improve predictive performance. The Linear Regression model achieved an RMSE of $2,016.61 and an R^2 of 0.5016.

Random Forest and Gradient Boosting were also evaluated because tree-based ensemble methods can capture more complex and nonlinear relationships in the data. Before hyperparameter tuning, Random Forest achieved an RMSE of $1,903.31 and an R^2 of 0.5560, while Gradient Boosting achieved an RMSE of $1,945.67 and an R^2 of 0.5361. These results showed that both ensemble methods improved upon the Linear Regression baseline, with Random Forest providing the strongest initial performance.

Hyperparameter tuning was then applied to the Gradient Boosting model to improve its predictive performance. After tuning, Gradient Boosting achieved an RMSE of $1,884.82, an MAE of $1,218.61, and an R^2 of 0.5646. Based on RMSE, the tuned Gradient Boosting model provided the best overall predictive performance among the models evaluated in this project. However, the analysis also showed that extreme high-cost claims remain more difficult to predict accurately because these claims represent only a very small portion of the dataset.

Based on these findings, the tuned Gradient Boosting model is most appropriate as a decision-support tool for ranking and prioritizing claims by predicted cost rather than for automatically setting claim reserves. Claims with higher predicted costs can be identified for earlier review, while experienced claims professionals remain responsible for evaluating individual cases and making final business decisions.

IV. Methodology / Approach

This project used 188,318 claims from the Allstate Claims Severity dataset.

Three models were tested: **Linear Regression**, **Random Forest**, and **Gradient Boosting**. After tuning, Gradient Boosting performed best with an RMSE of $1,884.82.

The final model can help rank high-cost claims for early review.

V. Data Analysis

The analysis showed that 83.59% of claims are below $5,000, while only 23 claims are above $40,000. This shows that very expensive claims are rare.

Three regression models were compared: Linear Regression, Random Forest, and Gradient Boosting. After tuning, Gradient Boosting performed best, with an RMSE of $1,884.82, an MAE of $1,218.61, and an R^2 of 0.5646.

The final model can help rank potentially high-cost claims for early review.

A. *Key Dataset Attributes*

- 188,318 insurance claims
- 116 categorical features
- 14 numerical features
- Target variable: *loss* (claim cost)

B. *Analysis Focus Areas*

- Understand the distribution of claim costs
- Identify potentially high-cost claims
- Compare Linear Regression, Random Forest, and Gradient Boosting
- Select the best-performing model
- Use predictions to rank claims for early review

VI. Expected Outcomes

The expected outcome is a machine-learning model that can predict and rank insurance claims by expected cost.

The tuned Gradient Boosting model can help:

- Identify potentially high-cost claims early
- Rank the top 5% of claims for early review
- Help claims managers prioritize their work
- Support better reserve planning
- Support claims professionals rather than make automatic reserve decisions

The top 5% is an initial pilot threshold and should be tested before full operational use.

VII. Timeline

The project was completed over eight weeks. Table I summarizes the major activities completed during each stage.

TABLE I
PROJECT TIMELINE

Week	Activities
Weeks 1–2	Selected the project topic and dataset
Weeks 3–4	Cleaned the data and performed exploratory data analysis (EDA)
Week 5	Built and evaluated regression models
Week 6	Compared models and performed hyperparameter tuning
Week 7	Completed final model evaluation and dashboard development
Week 8	Completed the final report, presentation, and project submission

The major project tasks included data preparation, exploratory data analysis, model development, model comparison, hyperparameter tuning, final evaluation, dashboard development, and preparation of the final report and presentation.

VIII. Conclusion

This project showed that machine learning can help predict and rank insurance claim costs. Among the models tested, the tuned Gradient Boosting model performed best, with an RMSE of $1,884.82, an MAE of $1,218.61, and an R^2 of 0.5646.

The final model should be used as a decision-support tool to help claims managers identify potentially high-cost claims earlier. The top 5% of model-scored claims can be routed to experienced adjusters for early review before reserves are finalized.

The 5% threshold should be treated as an initial pilot threshold and tested before full operational use. The model is intended to support claims professionals and improve decision-making, not replace professional judgment or automatically set claim reserves.

[1] [2] [3] [4] [5]

References

[1] DanaFerguson, Meg Risdal, NoTrick, Sara R, Sillah, Tim Emmerling, and Will Cukierski. Allstate claims severity. https://kaggle.com/competitions/allstate-claims-severity, 2016. Kaggle.

[2] Dana Ferguson, Meg Risdal, Sara R NoTrick, Tim Emmerling Sillah, and Will Cukierski. Allstate claims severity, 2016.

[3] Rajeeva Gaur, Jeff Pickelman, and Hongyi Wang. Allstate insurance claims severity: A machine learning approach.

[4] Stephen Klosterman. *Data Science Projects with Python: A case study approach to gaining valuable insights from real data with machine learning*. Packt Publishing Ltd, 2021.

[5] Anthony Y Stringer, Kathleen Fuchs, Veronica Bordes Edgar, Thomas Bristow, Stephen Correia, Suzanne Penna, Anny Reyes, Douglas M Whiteside, Cady Block, Douglas Bodin, et al. The minnesota conference proposed guidelines for education and training in clinical neuropsychology. *The clinical neuropsychologist*, 40(2):331–375, 2026.